

Maximal Rank Is Not Necessary for BBM Coercivity: A Rotating-Half-Space Counterexample

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Status and source question

Status: candidate full counterexample, likely valid pending expert review.

Gennaioli–Stefani [1] consider nonnegative locally integrable kernels ρ_t and the BBM functionals

$$\mathcal{F}_{t,p}^\rho(u) = \int_{\mathbb{R}^N} \frac{\|u(\cdot + z) - u\|_{L^p}^p}{|z|^p} \rho_t(z) dz, \quad 1 \leq p < \infty.$$

Their Theorem 1.2 says that their kernel hypotheses plus maximal rank imply coercivity: if $u_k \rightarrow u$ in L^p and $\liminf_k \mathcal{F}_{t_k,p}(u_k) < \infty$, then $u \in W^{1,p}$ for $p > 1$, and $u \in BV$ for $p = 1$. Immediately afterward, on source PDF page 3, they ask whether maximal rank is necessary for this conclusion.

In parts (A) and (B) of Theorem 1.1, the sequence $(t_k)_{k \in \mathbb{N}}$ is not necessarily the same. We refer to Remark 3.2 for an example where one must pass to a subsequence of $(t_k)_{k \in \mathbb{N}}$ in the implication (A) $\implies$ (B). Moreover, as observed in [51, 61], property (ii) in (B) can be further refined by additionally requiring that the family $(\rho_t)_{t \in I}$ has *maximal rank* (see Definition 2.8 below for the precise formulation).

Theorem 1.2. *Let $p \in [1, \infty)$. Assume that (A) or (B) holds and that $(\rho_{t_k})_{k \in \mathbb{N}}$ has maximal rank. If $(u_k)_{k \in \mathbb{N}} \subset L^p(\mathbb{R}^N)$ is such that $u_k \rightarrow u$ in $L^p(\mathbb{R}^N)$ as $k \rightarrow \infty$ for some $u \in L^p(\mathbb{R}^N)$ and $\liminf_{k \rightarrow \infty} \mathcal{F}_{t_k,p}(u_k) < \infty$, then $u \in \mathcal{S}^p(\mathbb{R}^N)$.*

When the conclusion of Theorem 1.2 holds—that is, when the finiteness of the $\liminf$ of the functionals along an L^p convergent sequence implies that the limit function belongs to some subspace $\mathcal{X}^p(\mathbb{R}^N)$ of $L^p(\mathbb{R}^N)$ (e.g., $\mathcal{X}^p(\mathbb{R}^N) = \mathcal{S}^p(\mathbb{R}^N)$)—we say that the functionals $(\mathcal{F}_{t,p})_{t \in I}$ are *coercive* on $\mathcal{X}^p(\mathbb{R}^N)$ (see Definition 2.5 below for a more precise statement).

Any radially symmetric family $(\rho_t)_{t \in I}$ has maximal rank, but non-radially symmetric families with maximal rank are known (examples can be found in [61]). We do not know if the maximal rank condition is also necessary for the conclusion of Theorem 1.2 to hold.

To prove Theorems 1.1 and 1.2, we mix the approaches of [34, 51, 61] in a new fashion.

Proof intuition

The functional cannot detect how mass is split between z and $-z$: its translation-difference factor is even. Begin with the normalized radial disk kernel, whose energies are coercive. At every scale, move all mass from one half-disk to the opposite half-disk and double what remains. This changes the raw directional mass, but not a single value of the functional. If the chosen half-disk rotates through a dense set of orientations as the scale shrinks, every fixed narrow cone is entirely missed infinitely often. Thus coercivity survives exactly while maximal rank fails as strongly as possible.

Counterexample theorem

Let $N = 2$, let $\omega_2 = |B_1|$, and for $0 < t < 1$ put

$$\begin{aligned}\kappa_t(z) &= \frac{\mathbf{1}_{B_t}(z)}{\omega_2 t^2}, & v_t &= (\cos(\log(1/t)), \sin(\log(1/t))), \\ \rho_t(z) &= 2\kappa_t(z)\mathbf{1}_{\{z \cdot v_t > 0\}}, & t_k &= e^{-k}.\end{aligned}$$

Theorem 1. *For every $p \in [1, \infty)$, the family (ρ_{t_k}) satisfies condition (A) of Theorem 1.1 in [1], and the functionals $(\mathcal{F}_{t_k, p}^\rho)$ are coercive on*

$$\mathcal{S}^p(\mathbb{R}^2) = \begin{cases} W^{1,p}(\mathbb{R}^2), & p > 1, \\ BV(\mathbb{R}^2), & p = 1. \end{cases}$$

Nevertheless (ρ_{t_k}) does not have maximal rank. Consequently, maximal rank is not necessary for the conclusion of source Theorem 1.2.

Proof. The half-disk has half the area of B_t , so $\rho_t \geq 0$, $\rho_t \in L^1(\mathbb{R}^2)$, and $\int \rho_t = 1$. Its support lies in B_t ; hence $\rho_{t_k} dz \rightharpoonup^* \delta_0$. Moreover, for every $R > 0$,

$$R^p \int_{\mathbb{R}^2} \frac{\rho_{t_k}(z)}{R^p + |z|^p} dz \leq \int_{\mathbb{R}^2} \rho_{t_k}(z) dz = 1.$$

This is condition (A), with $\alpha = 1$.

For $u \in L^p(\mathbb{R}^2)$ define

$$A_u(z) = \frac{\|u(\cdot + z) - u\|_{L^p}^p}{|z|^p}, \quad z \neq 0.$$

A change of variables shows $A_u(-z) = A_u(z)$. Since κ_t is even and the hyperplane $z \cdot v_t = 0$ is null,

$$\begin{aligned}\mathcal{F}_{t,p}^\rho(u) &= 2 \int_{z \cdot v_t > 0} A_u(z) \kappa_t(z) dz \\ &= \int_{z \cdot v_t > 0} A_u(z) \kappa_t(z) dz + \int_{z \cdot v_t < 0} A_u(z) \kappa_t(z) dz \\ &= \mathcal{F}_{t,p}^\kappa(u).\end{aligned}$$

The equality holds for every t , every p , and every u , including when the value is infinite.

The radial family (κ_t) has maximal rank and satisfies condition (A). Therefore Theorem 1.2 of [1] gives: whenever $u_k \rightarrow u$ in L^p and $\liminf_k \mathcal{F}_{t_k, p}^\kappa(u_k) < \infty$, one has $u \in \mathcal{S}^p(\mathbb{R}^2)$. The exact functional identity transfers this coercivity statement verbatim to (ρ_{t_k}) .

It remains to disprove maximal rank. Fix any $w \in \mathbb{S}^1$ and $\tau \in (0, 1)$. The cone

$$\mathcal{C}_\tau(w) = \{z : z \cdot w \geq (1 - \tau)|z|\}$$

has angular half-width $a_\tau = \arccos(1 - \tau) < \pi/2$. The sequence $v_{t_k} = (\cos k, \sin k)$ has dense tails on $\mathbb{S}^1$. Hence there are infinitely many k for which the angle between v_{t_k} and $-w$ is less than $\pi/2 - a_\tau$. For each such k ,

$$z \cdot v_{t_k} < 0 \quad \text{for every } z \in \mathcal{C}_\tau(w) \setminus \{0\},$$

and consequently, for every $\delta > 0$,

$$\int_{B_\delta \cap \mathcal{C}_\tau(w)} \rho_{t_k}(z) dz = 0.$$

Thus the liminf cone mass is zero for every fixed w and every admissible τ . No basis can satisfy Definition 2.8 of [1]; maximal rank fails. $\square$

Verification, limitations, and novelty

The proof uses no numerical approximation. The normalization, condition (A), exact even-part identity, coercivity transfer, and cone exclusion were checked separately; details are recorded in `verification.md`. Notice that no symmetry of ρ_t is among the source hypotheses. The construction answers the literal question, but it does not decide whether maximal rank (or a symmetrized variant) is necessary when every kernel is required to be even.

The run indexes and exact-phrase/arXiv searches through 11 August 2026 found the current source, whose November 2025 revision still asks the question, but no later resolution or matching rotating-half-space construction. Novelty confidence is moderate to high, subject to specialist citation review.

Human-review recommendation

Verify that Definition 2.8 is indeed applied to the raw kernel rather than its even part. If so, the counterexample is complete. A useful follow-up is to reformulate maximal rank invariantly in terms of $(\rho_t(z) + \rho_t(-z))/2$ and revisit necessity under that added symmetry.

References

- [1] L. Gennaioli and G. Stefani, *Sharp conditions for the BBM formula and asymptotics of heat content-type energies*, arXiv:2502.14655 (2025; revised 2025).